Synthetic Cultural Agents from Aggregate Anchors

A Falsifiable Protocol for Constructing Aggregate-Anchored Choice Policies

Augusto Gonzalez-Bonorino[1] Kseniia Biriukova[2] Monica Capra[1,3]

[1]EconLLM Lab and Arizona State University [2]Arizona State University [3]Claremont Graduate University

Working paper · August 2026

Corresponding author: Augusto Gonzalez-Bonorino, EconLLM Lab and Arizona State University. Email: agonz439@asu.edu.

Significance. Language models are increasingly used as stand-ins for survey populations, but it is rarely clear what information entered the model or what result would reject the construction. We specify a protocol that maps a declared country-level preference profile into a synthetic choice policy without putting a country name in the evaluation prompt, then test that policy on sixteen countries. Prompting recovers the shape of World Values Survey answers; only the trained weights recover the Global Preference Survey direction. The result is a usable, rejectable object—not a recovered culture, not a substitute for human data.

Abstract

Social scientists increasingly use language models to simulate survey respondents or to align models with named communities. Both uses leave unclear what information entered the model and what evidence would reject the construction. We declare an *aggregate anchor*—a vector of preexisting population-level measurements—and freeze a protocol that maps it into synthetic pairwise data and then into a language-model choice policy. Country names are absent from primary evaluation prompts. The object is an aggregate-anchor-induced synthetic choice policy, not an identified person and not a recovered human response distribution. We implement the protocol with Global Preference Survey country scores, sign-labeled synthetic pairs, and Direct Preference Optimization, and evaluate sixteen country adapters on World Values Survey Wave 7 against base-model, country-prompt, uniform-noise, and non-LLM baselines. Prompting recovers distributional shape (pooled TVD 0.375 vs. adapter 0.469 and noise 0.338). Only the trained weights recover the GPS trust direction (ρ = 0.78 vs. persona 0.06; human layer 0.41), and a restricted permutation placebo attributes that ranking to the real anchor assignment (0.782 vs. null 95th percentile 0.453, p = 0.001). A weights-by-prompt factorial shows the two channels compose. The protocol is useful where those tests pass and should be rejected where they fail.

Keywords: synthetic cultural agents; aggregate anchors; language models; preference optimization; survey simulation; falsification; culture

1. A construction problem, not an impersonation problem

Language models are now routinely asked to stand in for people: a prompt supplies a demographic profile, a country name, or a social identity, and the model is asked to answer a survey "as" that person or population. This practice is attractive because it is cheap and scalable. It is also scientifically precarious. Prompted synthetic respondents can be highly sensitive to wording and model version, and their distributions can differ materially from human survey distributions [1, 3, 9]. More fundamentally, an inference-time persona makes it difficult to distinguish information supplied by the researcher from stereotypes or associations retrieved by the model.

The alternative is not to claim that a model has discovered a culture. It is to construct a model object from declared population-level information and then submit that construction to tests that can fail. We call the resulting object an *aggregate-anchor-induced synthetic choice policy*. The policy is built from a country-level preference profile, but it is not asserted to represent any individual in that country. This boundary matters: evidence that models can flatten or misportray identity groups is a warning against treating synthetic outputs as voices of the people they purport to emulate [30].

Our contribution is a protocol, not a theorem of human-preference recovery. A protocol specifies: which aggregate measurements enter; how they are rendered; how scenarios, response pairs, labels, and quality filters are generated; how the policy is estimated; and which internal, placebo, external, and sensitivity tests decide whether the construction is useful. The protocol is constructive because it creates a policy under researcher-authored rules. It is falsifiable because each link in the construction makes a separate empirical commitment.

The first implementation uses the Global Preferences Survey (GPS), which reports country-level profiles for trust, risk taking, patience, altruism, and positive and negative reciprocity [14]. These anchors yield synthetic preference pairs; Direct Preference Optimization (DPO) fits lightweight country-specific adapters relative to a common base policy [24]. The primary external criterion is a theory-mapped battery of World Values Survey (WVS) item distributions. Crucially, country information is excluded from the primary evaluation prompt. This removes a direct country-token channel; it does not prove that all country-associated information is absent from base-model knowledge, scenario wording, or the profile. Thus the relevant question is not whether a model can retrieve a country label at inference. It is whether a policy induced by a declared aggregate anchor has externally useful behavior under a transparent, fixed procedure.

This paper makes four contributions: a formal statement of the aggregate-anchor construction problem and its scientific object; a modular protocol with anti-leakage, provenance, and reproducibility requirements; an empirically testable design built on five testable links; and a sixteen-country comparison against base-model, country-prompt, non-LLM, and permuted-anchor baselines, whose results we report without assuming transport.

2. Positioning: where this protocol differs

The closest literatures differ in both their information source and the location at which that information enters the model (Table 1). Early "silicon sample" work conditions models on detailed respondent profiles and asks whether generated responses resemble human samples [1, 2, 6]. Subsequent evidence finds instability, distributional mismatch, and inferential risks in such replacement exercises [3, 4, 9]. A second family fine-tunes models on human preference or survey-response data [28, 21, 5]. These approaches may be appropriate when individual records or empirical response distributions are available; their claim is correspondingly closer to learned human response behavior. A third family evaluates or aligns models against named cultural communities [29].

Table 1: Four approaches to LLM behavior associated with social-science targets. The final row is this paper's object.

Approach	**Training information**	**Conditioning locus**	**Central claim**
Persona prompting	identity or demographic profile	inference prompt	model imitates a named respondent/group
Microdata fine-tuning	individual responses or distributions	weights and/or prompt	model learns observed response behavior
Cultural alignment	named community preferences or norms	weights and/or prompt	model aligns to an identified target
Aggregate-anchor SCA	synthetic pairs induced by declared aggregate measurements	weights; no country prompt in primary evaluation	a protocol-induced policy has passed stated tests of transport and robustness

The protocol uses familiar computational components and makes no novelty claim for them: conditional probabilities from a language model [22], pairwise preference fitting via DPO [24], and low-rank adapters [19, 12]. The contribution is the specification of how aggregate measurements enter the synthetic-data operator and how the resulting policy is subjected to tests that distinguish construction success from merely plausible prose.

3. The scientific object and its boundary

3.1 Aggregate anchors

For country c, let

$$z_c = (z_{c1}, \dots, z_{cK}) \in \mathbb{R}^K \tag{1}$$

be a vector of pre-existing population-level measurements selected before training. In the first implementation, $K = 6$ and z_c contains GPS country scores, represented in the pipeline as standardized deviations from global means. An aggregate anchor is an input to a construction procedure. It is *not* an individual record, a country name shown at inference, or a moment condition automatically satisfied by a trained policy.

We call such an input a *declared* anchor: it is chosen by the researcher from a pre-existing, citable measurement source under a written inclusion rule, and it is fixed before training and before any outcome is inspected. A declared anchor is thus a researcher-committed input, whereas an *identified* anchor is one recovered from the data.

3.2 A frozen construction protocol

Figure 1 summarizes the object and its tests. We define a protocol as

$$\mathcal{P} = (Z, R, G, S, Q, A, E), \tag{2}$$

where Z records the anchor source and country frame; R is the deterministic anchor-to-profile rendering rule; G is the scenario, response-pair, and label-generation procedure; S is scoring, quality control, and filtering; Q records prompts and direct-token anti-leakage restrictions; A is the policy estimator and its training configuration; and E is the planned evaluation, placebo, and sensitivity design. Conditional on an archived run manifest, the model-mediated stages of G and S are stochastic realizations; the protocol is not equivalent to a deterministic mapping from z_c to data. In the implemented protocol, Z is the GPS ingestion and country frame of Section 6.1, R the deterministic profile renderer, and G the teacher–generator bank together with deterministic sign labeling (Section 4.2). S is inherited pair scoring used as a diagnostic, Q the frozen prompts, A the DPO fit of Section 4.3, and E the tests of Section 5.

Given a reference policy π_{ref}, the protocol generates a country-specific synthetic dataset $\mathcal{D}_c = \mathcal{G}_{\mathcal{P}}(z_c)$ and fits

$$\widehat{\pi}_c^{\mathcal{P}} = \mathcal{A}(\mathcal{D}_c; \pi_{\text{ref}}) \, . \tag{3}$$

Equation (3) defines a conditional distribution over candidate responses induced by a declared anchor and a frozen protocol. It does *not* assert $\widehat{\pi}_c^{\mathcal{P}} = P_{\text{human},c}$, where $P_{\text{human},c}$ denotes a human response distribution. That equality is an empirical transport hypothesis, not a definitional fact.

3.3 What this construction does not identify

Country averages do not identify which individuals hold which preferences, how preferences co-vary with demographic characteristics, or the full within-country distribution. The protocol therefore does not claim to recover individual heterogeneity, subgroup distributions, or a country's latent culture. Nor does it license reduced-form causal estimation, structural parameter recovery, or policy counterfactuals without a separate measurement and identification argument. These are not rhetorical caveats; they determine the object of study.

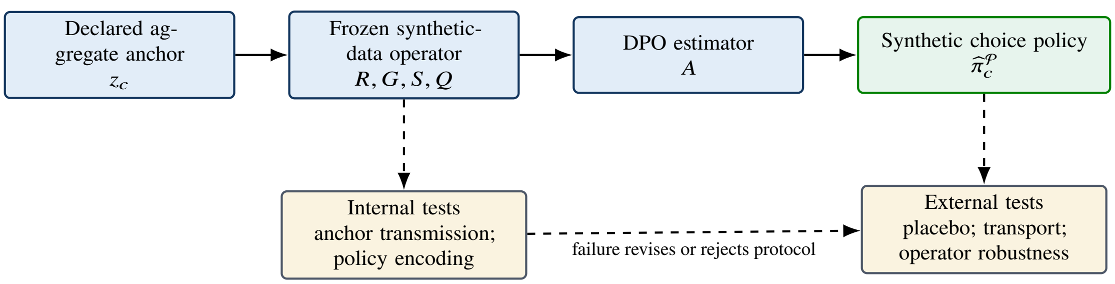


Figure 1: The paper's scientific object. Solid arrows construct a policy. Dashed arrows are tests, not guarantees. A failed test rejects a claim about this protocol; it does not prove that a population lacks the relevant preference.

4. From anchors to synthetic choice policies

4.1 Anti-leakage and profile rendering

The primary implementation begins with the GPS vector and renders it into an anonymized quantitative profile. The rendering states the magnitude and direction of each standardized score without inserting a country name. In the primary evaluation prompt, country and demographic labels are likewise omitted. This direct-token anti-leakage design does not establish cultural validity or prove that country-associated information is absent. It establishes a narrower, auditable fact: any difference between adapters cannot be attributed to country tokens in the primary inference prompt. The claim must be complemented by audits of scenario text, profile re-identifiability, and whether a classifier can infer country from generated artifacts.

4.2 Synthetic data operator

Figure 2 records the implemented path. The operator has two layers. A teacher first proposes behaviorally distinct facets and culturally neutral scenarios for each preference dimension, and a generator then writes two opposing responses to each scenario while holding non-target traits constant. Response A is instructed to load positively on the target dimension and Response B negatively. That shared bank does not depend on the country. Country identity enters only as a label: the production rule assigns the high-loading response when $z_{c,k} \geq 0$ and the low-loading response when $z_{c,k} < 0$. An independent scorer may assign both responses six-dimension scores; in the implemented protocol those scores are inherited from the materialized bank and used as diagnostics. Contamination is recorded and is not a retention gate. A legacy LLM selector that was prompted to implement the same sign rule is retained only as a sensitivity arm.

Example (illustrative). Mexico's GPS patience score is -0.11. The rendering stage states that standardized patience is moderately below the global mean. The teacher proposes a culturally neutral intertemporal scenario, and the generator fixes two opposing responses: A waits for the larger later reward (positive loading) and B takes the smaller reward now (negative loading), with risk attitudes and income constraints held as constant as the prompt can make them. The sign rule, seeing a negative z, labels B as chosen for Mexico. Inherited scorer ratings, when available, are attached as diagnostics and do not decide admission. This trace illustrates the implemented mechanics; it is not an empirical result.

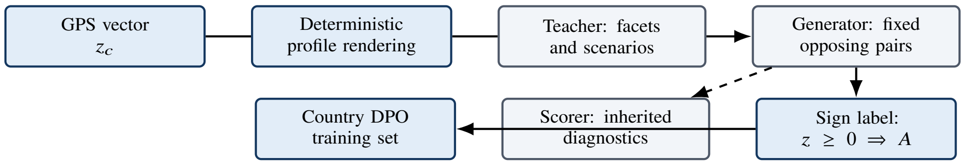


Figure 2: Implemented architecture. Profile rendering and sign labeling are deterministic; the teacher and pair generator are the stochastic bank. The scorer is diagnostic and is not a contamination gate. An LLM selector is a sensitivity arm, not this path.

This decomposition matters. Calling every upstream choice "teacher bias" obscures the actual sources of dependence: scenario construction, response-pair generation, the labeling rule, scorer judgments, and any filter thresholds may each change the synthetic training distribution. The empirical design therefore treats the full operator as an object of sensitivity analysis.

4.3 Policy estimation

For synthetic triplets $(x, y_w, y_\ell) \in \mathcal{D}_c$, DPO fits a policy relative to π_{ref} by minimizing

$$\mathcal{L}_{\text{DPO}}(\pi) = -\mathbb{E}_{(x,y_w,y_\ell)\sim\mathcal{D}_c}\left[\log\sigma\left(\beta\log\frac{\pi(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta\log\frac{\pi(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)}\right)\right]. \tag{4}$$

Equation (4) is a pairwise-preference estimator with reference-policy regularization [24]. Here it fits synthetic preferences generated by $\mathcal{P}$. It is not a constrained estimator that mechanically enforces $\mathbb{E}[f(Y)] = z_c$, and we make no such claim. Intensity-weighted pairwise variants that down-weight near-zero anchors are natural sensitivity arms in E [26].

5. Making the construction empirically rejectable

The phrase "falsifiable" is useful only if it names propositions, estimands, and failure consequences [11, 8]. This paper is not a public preregistration. The archived analysis plan fixed the unit of analysis, primary metric, comparison, practical margin, randomization count, uncertainty procedure, and decision rule for every row in Table 2 before outcomes were inspected. The five links have been exercised by the sixteen-country evaluation; per-link status is in Table 2. Each test concerns a distinct link; passing one does not imply passing another.

Table 2: The five rejection points of an aggregate-anchor construction.

Claim	Predeclared evidence	Status after the sixteen-country evaluation	Interpretation of failure
Anchor transmission	label/sign agreement, target separation, QC rates, anchor swaps	PASSED — deterministic sign labeling ($G = \text{sign}(z)$) is the default; label/sign agreement verified on the production 658-pair bank	the synthetic operator does not express the declared input reliably
Policy encoding	held-out synthetic-pair recovery; policy-distance stability	PASSED — held-out synthetic-pair accuracy 0.886–0.992 ($n = 132$ per country)	DPO does not retain the contrast present in the synthetic data
Anchor relevance	real-anchor statistic versus a permuted-anchor null	PASSED — restricted permutation placebo (N=1000, seed 20260820): real trust bridge $\rho = 0.782$ vs. null median -0.006 and 95th percentile 0.453; $p = 0.001$ one-sided. Secondary rank statistic degenerate by construction (Section 5.2)	apparent gains may reflect generic synthetic tuning rather than anchor information
External transport	independent human survey distributions versus base, prompt, and non-LLM baselines	PARTITIONED — distributional shape: adapters (TVD 0.469) lose to the persona prompt (0.375) and uniform noise (0.338); construct bridge: adapter trust $\rho = 0.78$ vs. persona 0.06 (human layer 0.41), development partials +0.50 (trust) / +0.46 (patience). Verdict: no histogram reproduction; preference direction and its economic gradient transport	this protocol does not generalize to the chosen external criterion
Operator robustness	scenario, generator, selector, scorer/QC, materialized-data, and training-seed ablations	PARTIAL — inference-time temperature scaling measured (pooled TVD 0.469 $\to$ 0.332 at $T = 3$, top-option match unchanged); weights-by-prompt interaction measured (bridge ρ: adapter-persona 0.60, adapter-uncond 0.78, base-persona 0.06; shape unchanged) — direction survives prompting; training-side ablations pending (extended-prompt bank, soft-DPO/$\lvert z \rvert$ pruning, repeated seeds, deterministic-vs-legacy selector)	the conclusion depends on arbitrary or unstable upstream choices

5.1 Internal tests: transmission and encoding

Anchor transmission asks whether changing z_c changes labels and retained pairs in the direction dictated by the declared rule. It is assessed with country-by-dimension label summaries, quality-control pass rates, target-versus-nontarget separation, contamination ratios, and anchor-swap diagnostics. The current implementation makes an especially important test available: because the selector is instructed to implement a sign rule, the primary sensitivity comparison must contrast a deterministic sign-labeling operator with the legacy LLM selector. Policy encoding asks whether the trained adapter reproduces held-out synthetic contrasts and whether independently trained policies are stable under repeated materializations of the same protocol.

5.2 Anchor relevance: a restricted permutation distribution

A central negative control evaluates the real anchor assignment against a null in which anchors are permuted while all other stages are retained. The implemented placebo is deliberately restricted: rather than retraining placebo adapters under permuted anchors, it draws N=1000 restricted permutations of the sixteen country-to-anchor assignments (drawn without replacement from the observed anchors, seed 20260820) and recomputes the predeclared primary statistic — the trust construct-bridge Spearman ρ between the adapter composite and the GPS country z-score — on each permutation. Two statistics are declared non-computable under the restriction rather than silently omitted: the median own-country rank is invariant under this null by construction (sixteen countries instantiate only thirteen sign profiles, so permuting the observed anchors never changes the set of profile-adapters), and the development-restriction correlation was not precomputed per permutation and is reported as descriptive context only.

The real assignment attains $\rho = 0.782$ against a null median of -0.006 and a 95th percentile of 0.453: $p = 0.001$ (one-sided), so anchor relevance passes on the predeclared primary statistic. The rank degeneracy is itself a finding: sign-profile collapse makes location-rank statistics insensitive to anchor permutation, which identifies the bridge — a magnitude-based statistic — as the discriminating test. The restriction also bounds the claim: the placebo establishes that the realized anchor assignment carries information beyond a generic synthetic-tuning effect at the level of the trained bank's bridge; it does not establish that adapters retrained under permuted anchors would fail identically, and the full retraining placebo remains part of the protocol's rejection design.

5.3 External transport and uncertainty

External transport is a predictive-criterion claim: the policy's induced option probabilities improve alignment with independently collected human response distributions. It is neither recovery nor validation of a complete country distribution. For a question q with valid options $\mathcal{Y}_q$, let $\ell_{\widehat{\pi}_c}(y \mid q)$ denote the total log probability assigned to the token sequence encoding answer option y under a fixed prompt and option-order rule. The policy yields the option probabilities in equation (5),

$$\widehat{P}_c(y \mid q) = \frac{\exp\{\ell_{\widehat{\pi}_c}(y \mid q)\}}{\sum_{j \in \mathcal{Y}_q} \exp\{\ell_{\widehat{\pi}_c}(j \mid q)\}}, \tag{5}$$

and the primary descriptive discrepancy is the total variation distance in equation (6) (with frequency-based fixation indices [23] as a dispersion-sensitive alternative),

$$\mathrm{TVD}(\widehat{P}_c, P_c) = \frac{1}{2} \sum_{y \in \mathcal{Y}_q} |\widehat{P}_c(y \mid q) - P_c(y \mid q)|. \tag{6}$$

For each country–item cell, P_c is the survey-weighted empirical option distribution after a frozen harmonization map for option wording, order, invalid outputs, and Likert categories. We report absolute TVD and paired differences from every baseline. Finite-survey sampling, item aggregation, and training-seed uncertainty are distinguished where the design identifies them; a question bootstrap is only a robustness check on this item battery. A bootstrap over survey items is useful only as a robustness calculation for the declared item battery—not as respondent sampling uncertainty, an item-universe inference, or uncertainty about an upstream model-mediated pipeline.

6. Empirical implementation: predeclared design

6.1 Data roles and country frame

The implementation uses GPS as the anchor source and WVS Wave 7 as the primary independent outcome surface. The country frame is the sixteen-country panel reported in Section 7, drawn from the GPS×WVS intersection before outcomes were inspected. The evaluation battery is the theory-led map of thirty single-choice items plus three demographic items; confirmatory claims are restricted to trust and, directionally, patience. The remaining dimensions are reported as imprint diagnostics, not as recovered survey constructs.

AmericasBarometer and the Integrated Values Surveys may serve as secondary instrument or temporal replications if multi-country overlap is adequate. World Bank indicators are contextual covariates for exploratory heterogeneity analysis, not direct choice-distribution criteria.

6.2 Anchor sources: criteria, GPS, and alternatives

A source must satisfy a small set of conditions before it can anchor the construction. It must be pre-existing and citable, fixing the reference in the public record rather than constructing it for this study; it must offer population-level aggregates rather than individual microdata, because the outcome surface is itself country-level; and it must be theoretically linked to the constructs of interest while remaining independent of the evaluation surface, so that anchoring on WVS country means while evaluating WVS items would be circular. It must also offer broad country coverage and valid, comparable measurement across countries.

GPS is a defensible first implementation of these criteria. Its country scores are aggregates by construction, derived from measures of six economic preferences that were validated ex ante in incentivized experiments, collected in representative samples across 76 countries—a set that overlaps the WVS Wave 7 country frame. The data are public and citable, and the criteria are fixed before any scaled run.

Alternatives exist if the criteria recommend them: WVS/EVS country means, with the caveat of overlap with typical evaluation surfaces; Schwartz value country scores (PVQ/ESS) [25]; Hofstede dimensions [15]; Inglehart–Welzel cultural-map coordinates [20]; and the GLOBE project [13]. Table 3 lists these candidates with their principal caveats. The method is agnostic to the specific source, so long as the criteria above hold.

Table 3: Candidate anchor sources and their principal caveats.

Source	What it measures	Key caveat
GPS	Six economic preferences (trust, risk-taking, patience, altruism, positive and negative reciprocity)	Aggregate by construction; experimentally validated
WVS/EVS means	Country-level value orientations and attitudes	Circular when the evaluation surface is WVS itself
Schwartz (PVQ/ESS)	Ten basic human values at country level	Fewer countries; values are not preference measures
Hofstede dimensions	Work-related cultural dimensions	Dated samples; dimensions are not preference measures
Inglehart–Welzel map	Emancipative and secular value coordinates	Two-dimensional reduction of a larger item space
GLOBE	Societal practices and leadership values	Organizational focus; limited item overlap

6.3 Baselines and primary comparison

Every country is evaluated against four mandatory baselines: (i) the unmodified base model; (ii) the persona-prompt baseline (a country-named cultural prompt, deliberately violating direct-token anti-leakage to answer a distinct practical question: whether an explicit persona retrieves useful information more effectively than the constructed policy); (iii) a named non-LLM aggregate benchmark fixed before outcome inspection; and (iv) the restricted permuted-anchor distribution.

The named non-LLM benchmark is the maximum-entropy anchor transform: for each country–item cell, the anchor's standardized score $z_{c,k}$ is mapped to a target mean on the item's recoded option grid, $\mu = \bar{r}_q + \beta z_{c,k}$ with $\beta = 0.5$ declared before outcome inspection, and the option distribution is the maximum entropy distribution on that grid subject to the mean constraint (equivalently a single raking step from a uniform seed; [17, 16, 18]). Infeasible cells (target mean outside the grid) are clamped to the grid boundary, which degenerates to a point mass. A second arm realizes the training signal mechanically: all mass on the anchor-sign option ($z_{c,k} \geq 0 \Rightarrow$ high-loading option), the sign-follower. The baseline consumes exactly the adapters' information budget — declared country-level anchors, no WVS outcome data, no microdata, no LLM — so the comparison asks whether the LLM-DPO machinery adds anything over a mechanical transformation of the same anchors. The analysis must report provenance-sensitive construction success separately from predictive alignment; a cultural prompt may be practically predictive while failing the former objective.

The primary cross-country comparison is each fixed policy against each country's human distribution. Under unconditioned prompts this is *fixed-policy proximity*, not evidence that the model conditions on a country at inference. We report gains and failures separately rather than defining success as a uniformly positive average.

6.4 Operator-sensitivity remainder

Temperature scaling and the weights-by-prompt factorial are reported in Section 7. Training-side ablations—alternate materialized banks, deterministic-sign versus legacy-selector labels, repeated DPO seeds, and intensity-weighted losses—remain unrun. They belong to the rejection design; they are not results. The predeclared panel is archived in Appendix C.

7. Results: sixteen countries, four surfaces

All numbers in this section were re-derived from the frozen evaluation artifacts by the committed pipeline `analysis/phase2/` (G0 freeze, E[recode]/P(option 1), 2026-08-23); none are quoted from secondary summaries. Sixteen country-specific adapters — CHN, JPN, GBR, USA, MEX, ARG, DEU, RUS (first bank) and IND, IDN, NGA, EGY, TUR, NLD, BRA, GRC (second bank) — were trained on 526 synthetic sign-labeled pairs drawn from the shared 658-pair bank ($G = \text{sign}(z)$, $\beta = 0.1$, LoRA $r = 16/\alpha = 32$, one epoch; Appendix B) and evaluated on the WVS Wave-7 surface: 30 single-choice items (binary and ordered scales) mapped to the six GPS dimensions through the theory-led map. The single-choice surface follows the evaluation notebook's construction: the mapped items less the multi-select childrearing battery (Q12–Q13–Q14), plus the three single-choice demographic items (Q260, Q275, Q288); the multi-select and numeric-open items are excluded because their option space is not a single softmax. Four model families are compared on identical surfaces: the adapters, the unconditioned base, the base prompted as a resident of each country (persona), and uniform noise; the survey-weighted human distributions are the target.

7.1 Distributional fit: prompting wins, but the population is diffuse

Table 4 reports the pooled distributional surface (means over item×country cells).

Table 4: Distributional surface, pooled across 16 countries and 30 single-choice items. Lower TVD/JSD and smaller |entropy err|/|std err| are better; top-match is a rate. Base is one fixed distribution replicated across countries.

Model	TVD	JSD	Entropy err	Std err	Top-match
adapter	0.469	0.279	−0.737	−0.647	0.452
base	0.443	0.249	−0.540	−0.535	0.452
persona	0.375	0.188	−0.438	−0.449	0.479
noise	0.338	0.145	+0.568	+0.845	0.331

The first result is a floor, not a model. Uniform noise achieves TVD 0.338 — *below* every model — because the WVS populations are diffuse: on many items the population is near-uniform, so any concentrated distribution pays a TVD penalty and TVD rewards flatness as much as accuracy. Against this floor, the persona prompt is the best model family: pooled TVD 0.375, only 0.037 above noise, with the smallest under-dispersion (entropy error −0.44) and the best top-option match (0.48). The persona beats the adapters in 15 of 16 countries and the base in 15 of 16.

Why do the adapters underperform on TVD? They are the most over-concentrated family (entropy error −0.74, standard-deviation error −0.65), and they beat the base in only 5 of 16 countries. This is a direct consequence of hard pairwise DPO labels: a pairwise preference signal identifies an ordering, never population marginals. It replicates the mode-collapse diagnosis of Dalloul & Pfeifer (2026) on the same base model and of the wider synthetic-survey literature. Distributional shape is precisely what the weights-only construction does not deliver.

7.2 The construct bridge: direction lives in the weights

Table 5 reports Spearman correlations between country composites and GPS country z-scores, per dimension and family.

Table 5: Construct bridge: Spearman ρ of the country composite vs the GPS country z-score. Human layer $n = 42$; adapter and persona layers $n = 16$. Base and noise are single fixed distributions with no cross-country variation (undefined by construction).

Dimension	Human (42)	Adapter (16)	Persona (16)
patience	0.32	0.42	0.60
risktaking	−0.08	0.57	−0.11
posrecip	−0.00	0.48	0.02
negrecip	0.42	0.06	0.34
altruism	0.03	0.24	−0.23
trust	0.41	0.78	0.06

The key contrast is on trust, the strongest GPS proxy in this menu (the published country-level correlation is 0.49 [14]). The adapter composite ranks countries in GPS order ($\rho = 0.78$)[1]; the human WVS layer does too (0.41); the persona prompt does *not* (0.06). On patience the dissociation is dimension-specific: the adapter tracks (0.42) and so does the persona (0.60), against a human layer of 0.32 — the persona's patience ordering is plausibly inherited from development-correlated priors (Section 7.3). The remaining dimensions show the *imprint* pattern: risktaking (0.57), positive reciprocity (0.48), and altruism (0.24) show adapter correlations far *above* the human layer's (−0.08, −0.00, 0.03). Because those WVS items do not track GPS in humans, a high adapter ρ there is the GPS sign bit expressed through a poorly mapped instrument — an imprint, not a recovered survey construct, and a reminder that cross-national item comparability is the precondition for construct interpretation [7, 27, 10]. Figure 3 displays the bridge across all six dimensions, with base and noise undefined by construction.

[1]Composites are $\mathbb{E}[\mathrm{recode}(V)]$: binary items (Q57 and the child-quality dummies) enter as $P(\text{option} = 1)$; Likert inversions are linear and commute with the expectation. Missing codes are masked before recoding. Applying the nonlinear recode to the raw mean instead zeros Q57 in every country and yields adapter trust 0.70; a majority-threshold recode of the mean yields 0.79. The $\mathbb{E}[\mathrm{recode}]$ construction is reported throughout.

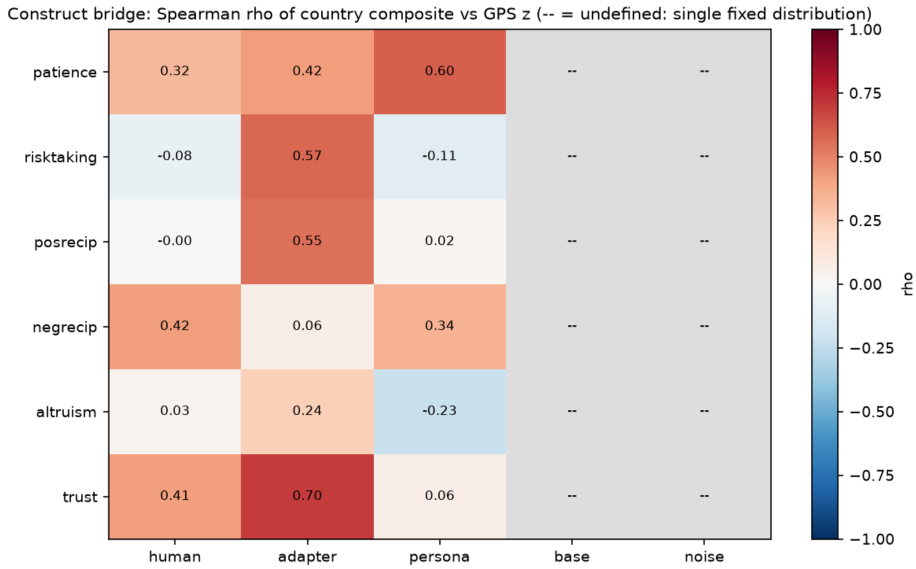


Figure 3: Construct bridge: Spearman ρ of the country composite versus the GPS country z-score, by dimension and model family. Gray cells are undefined (base and noise are single fixed distributions with no cross-country composite variation). The trust row is the paper's positive result: adapter 0.78, human layer 0.41, persona prompt 0.06. Human layer $n = 42$; adapter and persona layers $n = 16$.

7.3 Development restrictions: the adapters carry the anchor's gradient

Table 6 reports Spearman correlations of each composite with log GDP per capita (WDI 2015–2019), with education partials.

Table 6: Development restrictions: Spearman ρ of the country composite vs log GDP per capita (WDI 2015–2019); the adapter column also reports the partial controlling education (WVS Q275). Base and noise undefined by construction.

Dimension	Human (42)	Adapter (16)	Adapter partial|edu	Persona (16)
patience	0.09	0.29	0.46	0.33
risktaking	−0.34	0.35	0.33	−0.62
posrecip	−0.22	0.03	0.46	0.63
negrecip	0.42	−0.32	−0.49	−0.59
altruism	−0.41	0.27	0.54	0.50
trust	0.09	0.13	0.50	0.73

The adapters reproduce the GPS development gradient on the confirmatory dimensions: trust +0.13 (education partial +0.50) and patience +0.29 (+0.46), in the same direction as the GPS anchor itself (patience +0.58, trust +0.32 against log GDP per capita over the 76-country bank; 75 countries have WDI GDP coverage) and consistent with the human WVS layer (trust +0.09). The persona *overshoots* (trust +0.73), consistent with prompting inheriting the base model's Western-coded priors. The base and noise, being single fixed distributions, carry no country variation and are undefined on this surface by construction.

7.4 Encoding and direction

The internal encoding link passes: held-out synthetic-pair accuracy is 0.886–0.992 across the sixteen countries ($n = 132$ per country; 0.89–0.99 rounded), so the adapters retain the contrasts present in their training data. Table 7 reports top-option match by dimension. The adapters' modal answers match the population mode on positive reciprocity (0.78) and trust (0.52). The noise baseline reaches 1.00 on altruism and 0.77 on negative reciprocity purely because those populations are near-unimodal — top-option match alone is not a validity statistic.

Table 7: Top-option match rate by dimension and model. Noise reaches 1.00 on altruism and 0.77 on negative reciprocity only because those populations are near-unimodal.

Dimension	Adapter	Base	Persona	Noise
patience	0.16	0.03	0.03	0.12
risktaking	0.28	0.27	0.08	0.48
posrecip	0.78	0.87	0.87	0.10
negrecip	0.50	0.62	0.64	0.77
altruism	0.54	0.33	0.81	1.00
trust	0.52	0.52	0.57	0.15

7.5 Cultural distance: two geometries of (non-)specificity

Cultural distance is not a single quantity, and the two geometries we compute disagree in an instructive way. On the frequency-based F_{ST} geometry of [23] — which treats questions as loci and response distributions as alleles — the median rank of an adapter's own country among the 42 WVS populations is 20.5, indistinguishable from chance (21.5); 11 of 16 adapters are nearest to Canada, a mid-scale reference, and the best adapters (GBR rank 2, RUS rank 4, BRA rank 8) sit far from their own countries. On the location geometry — the pooled TVD of each adapter's fixed output distribution against all 42 populations — the median own-country rank is 15.0 (chance 21.5): weakly but systematically below chance, with GBR, GRC and USA each at rank 2, and no adapter closest to its own country (Supplementary Figure S7). The two metrics thus disagree about how much location information survives, and the disagreement is informative rather than contradictory: F_{ST} is a variance-ratio that penalizes the adapters' over-concentration, while TVD reads the location shift the anchors induced. The PCA of the mapped items tells the same story as the frequency geometry: adapter and persona positions track the model family, not the target country (Figure 4, right).

The nearest-country structure is dominated by a mid-scale attractor: Canada is the nearest of the 42 populations for 11 of 16 adapters on both geometries. This is not evidence of a Canadian cultural affinity in the adapters — Canada sits near the cross-country center of the item space, so it is the closest point for any distribution whose location signal is weak. The adapters whose own-country fit is worst are precisely the sign-twin / development-shadow set from the construct analysis: NGA (rank 28), IDN (31), MEX (31), EGY (35) and IND (40) all fall below chance, and their GPS profiles are nearly collinear with GDP.

The coarse structure that does survive is the sign profile. The 76-country GPS bank collapses to 35 sign(z) profiles (22 multi-country classes), and the sixteen adapters instantiate 13 distinct profiles (MEX≡RUS, IND≡GRC, IDN≡EGY). Sign-twin proximity on human WVS is explained by shared GDP (partial correlation with the GDP gap -0.09): country-level sign patterns are a development shadow, not cultural families, and the adapters inherit this structure by construction because the anchor enters through its sign.

What, then, do these objects carry of the countries that anchor them? The imprint pattern of the construct-bridge analysis is the answer in miniature. On dimensions where the WVS instrument tracks GPS in humans (trust, patience), the adapters' country ordering aligns with the anchor ($\rho = 0.78$ on trust) and the development gradient survives in sign; on dimensions where the instrument does not (risktaking, positive reciprocity, altruism), the adapters still express the anchor's sign bit, but through a poorly mapped instrument — an imprint, not recovered culture. The adapters are therefore identified for the anchor's aggregate preference ordering, not for country-specific dispersion, magnitudes, or latent cultural structure. Prompting buys the complementary half of the dissociation — distributional shape without ordering. Neither route reproduces a country's culture; each carries a provable fragment of it, and this section has measured which fragment.

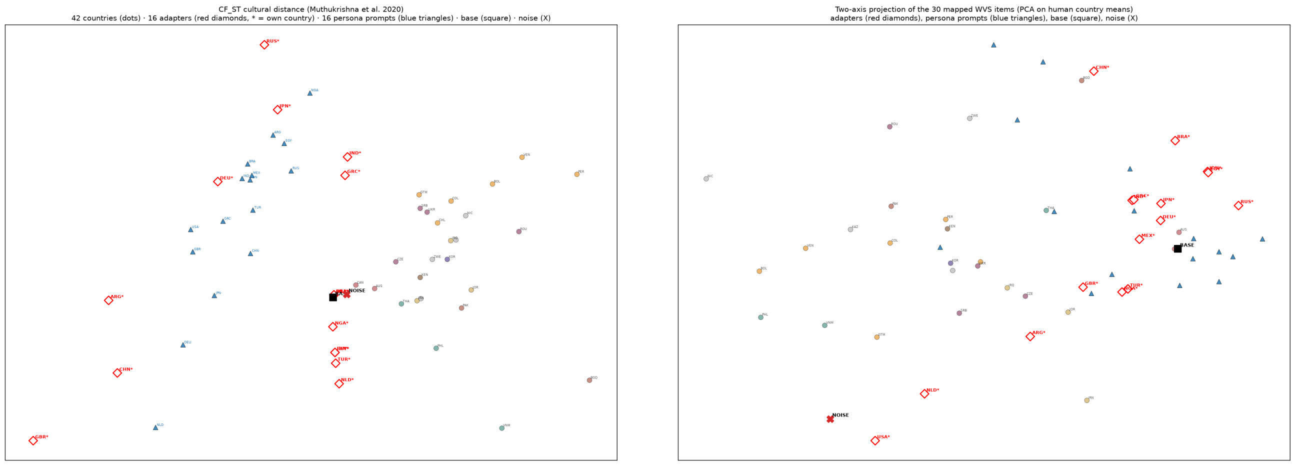


Figure 4: CF_ST cultural distance (left; Muthukrishna et al. 2020) and two-axis PCA of the mapped items (right). Red diamonds are adapters ($*$ = own country), blue triangles persona prompts, the square the base, $\times$ noise; country dots colored by region. Adapters do not localize near their own countries on either geometry.

7.6 The dissociation

The results dissociate the two routes into the model. The country-named prompt buys distributional shape (TVD 0.375, best dispersion and top-match) but carries no GPS ordering on trust ($\rho = 0.06$) and overshoots the development gradient (+0.73). The trained weights carry the GPS direction — trust bridge $\rho = 0.78$, development partials +0.50/+0.46, held-out pair accuracy up to 0.99 — but pay for it in concentration and in TVD (an anti-leakage price of ≈ 0.09; Table 4). The boundary of Section 3.3 applies verbatim: no histogram reproduction, no country-specific dispersion, no magnitudes. The dissociation also survives its negative control: under the restricted permutation placebo (Section 5.2), the trust bridge that anchors the weights-side of this reading ($\rho = 0.782$ real) exceeds the 95th percentile of the permuted-anchor null (0.453; $p = 0.001$), while no location-rank statistic can move under the null at all — so it is specifically the bridge evidence, not generic adapter quality, that carries anchor information.

7.7 The 2×2: weights and prompts compose

The four cells {base, adapter} $\times$ {unconditioned, persona} separate the two channels directly. Each country adapter was re-evaluated under the same country-named persona prompt used for the prompt baseline (sixteen countries, one seed, single prompt phrasing; all sixteen CSVs integrity-checked against the survey specification). On the trust bridge, the combination retains most of the weights' directional signal: adapter-under-persona reaches $\rho = 0.60$, against 0.78 for the adapters under their training-time unconditioned prompts and 0.06 for the base model under the same persona prompt — about 74% of the weights-versus-prompt bridge gap survives prompting. On shape, the trained weights cost nothing: pooled TVD under the persona prompt is indistinguishable between the adapter and the base (0.411 vs. 0.413 on this evaluation surface), with a slightly better top-option match (0.450 vs. 0.437).[2]

The interaction is therefore complementary rather than compensating: prompting supplies distributional shape, the trained weights supply GPS direction, and the two compose with modest attenuation of direction and no loss of shape. This reading strengthens, rather than softens, the protocol's boundary advice. The combined policy scores well on both surfaces — which is precisely why it must not be adopted uncritically: once a country name enters the prompt, anchor-derived direction and prompt-elicited stereotype priors are confounded in the output, and separating them again requires exactly this factorial decomposition. Provenance-clean construction keeps the channels separate by design; the 2×2 quantifies what re-mixing them does.

7.8 Operator diagnostics and the dispersion bound

Temperature scaling is an inference-time diagnostic. Raising the sampling temperature to $T = 3$ moves pooled TVD from 0.469 to 0.332 while the top-option match is unchanged (0.452): it recovers dispersion but does not move the location of the induced distribution. The shape loss is therefore a softmax artifact of low-temperature sampling, whereas the location loss is structural. The intensity-weighted training analogue of this lever is studied by [26].

The training-side levers were predeclared in Table 2. An extended-prompt training bank — richer completions, not binary A/B — is the only training-side lever that can change the shape of the learned distribution without touching WVS. Soft DPO

[2]Surface note: these TVD values are computed on this run's item universe and are not comparable to the canonical Table 1 numbers (0.375/0.469); comparisons are within-table. The bridge is unaffected by the surface change.

or $|z|$-pruning of near-zero anchors is a cheap ablation for knife-edge dimensions (patience, risktaking); we expect it to affect direction, not dispersion.

An explicit anti-contamination commitment: we will not train on population marginals. KL-to-wvs or moment-matching losses would contaminate the out-of-sample design, in which wvs stays unseen. Temperature scaling is reported as a diagnostic, not as a training change.

7.9 The non-LLM anchor transform

Table 8 reports the maximum-entropy anchor transform (armA) and the sign-follower (armB) on the same like-for-like surface: 27 mapped single-choice items (the multi-select battery excluded, as in the canonical construction), pooled over the sixteen countries.

Table 8: Like-for-like pooled distributional surface (27 mapped single-choice items, 16 countries). armA is the maximum-entropy anchor transform ($\beta = 0.5$); armB is the sign-follower. Lower TVD/JSD and smaller |entropy err| are better; top-match is a rate.

Model	TVD	JSD	Entropy err	Std err	Top-match
noise	0.349	0.147	+0.587	+0.371	0.381
armA	0.351	0.150	+0.568	+0.363	0.292
persona	0.366	0.179	−0.395	−0.418	0.514
base	0.444	0.243	−0.491	−0.501	0.467
adapter	0.470	0.278	−0.702	−0.629	0.476
armB	0.717	0.588	−1.722	−1.247	0.292

The anchor transform is the flatness floor with an anchor-induced location: its TVD (0.351) sits just above uniform noise (0.349) and below every LLM family. This is by construction — maximum entropy against a mean constraint is near-uniform on short grids, and TVD rewards flatness — so the comparison decomposes the adapters' histogram failure into a location component and a concentration component. The adapters carry the location signal (their modal answers match the population mode on positive reciprocity and trust at higher rates than the anchor transform, 0.78/0.52 versus 0.29 on both for armA) but pay a concentration penalty that the mechanical transform does not; the sign-follower armB is the worst construction on the surface (0.717), confirming that a sign-only signal without any intensity is the least useful anchor realization. On the anchor's own geometry the transform is trivially faithful by construction (its country composites correlate nearly perfectly with the anchors, $\rho \approx 1$), which is why the construct-bridge test is informative only for the trained adapters: $\rho = 0.78$ on trust is learned, whereas the transform's is mechanical. The 42-country location geometry ranks armA at the flatness attractor (median own-country rank 27.5 of 42, nearest population Kenya for all sixteen) and armB just below chance (14.0); neither arm recovers its own country (Supplementary Figure S9).

8. Discussion

The sixteen-country evaluation answers a narrower question than the title invites. A declared GPS sign bit, written into adapter weights with no country token at inference, ranks countries on trust the way GPS does ($\rho = 0.78$). A country-named prompt does not ($\rho = 0.06$). Uniform noise beats every model family on total variation because the survey populations are diffuse. Those three facts are the result.

They survive the tests that were run. The restricted permutation placebo attributes the trust ranking to the real anchor assignment ($p = 0.001$), not to generic synthetic tuning, but it does not retrain adapters and cannot move a location-rank statistic once sixteen countries collapse to thirteen sign profiles. The weights-by-prompt factorial shows the two channels compose: direction mostly survives a persona prompt ($\rho = 0.60$), and shape is unchanged. The non-LLM maximum-entropy transform sits on the flatness floor; it is faithful to the anchors by construction and therefore not a competitor on the bridge. Cultural-distance geometries disagree in an instructive way—frequency-based F_{ST} is chance, location TVD is weakly below chance—because the adapters carry ordering, not country-specific dispersion.

What the construction does not do is equally sharp. It does not reproduce histograms. It does not recover magnitudes or within-country heterogeneity. It does not license treating sixteen adapters as sixteen cultures. Training-side operator ablations remain unrun; until they exist, the direction result is a property of this materialized bank, not of every protocol in the family. The useful applications are those that need a transparent, rejectable preference *direction*—item design, benchmark construction, exploratory measurement. The method is not a substitute for human data and should not be used to speak for individuals, subgroups, or policy counterfactuals.

Code, data, and disclosures

Analysis code and the committed table artifacts live at `https://github.com/EconLLM-Lab/SCA2_PofW` (commit `a336ac0` and descendants). Headline numbers in Section 7 are regenerated by `analysis/phase2/13_unified_comparison.py`, `17_anchor_permutation_placebo.py`, and `18_2x2_analysis.py` from banked option-probability files. Those raw evaluation banks are not in the public tree; they will be deposited with a DOI before journal submission. Adapter weights are on the Hugging Face hub (`Bonorinoa/SCA2-phase2-adapters`). GPS data are documented by Falk et al. [14]; WVS Wave 7 data are available from the World Values Survey Association under DOI `https://doi.org/10.14281/18241.18` [31]. Raw survey microdata are not redistributed.

Preregistration. This study was not preregistered. The analysis plan was written before outcome inspection but was not posted to a public registry.

Funding. The authors declare no funding.

Generative AI. Language models are the scientific object of the study (synthetic-pair generation and the evaluated policies). Generative AI was also used as a writing and coding assistant. The authors take responsibility for all claims and numbers.

Competing interests. The authors declare no competing interests.

A. The two-country pilot: a fixed-policy proximity diagnostic

This pilot predates the sixteen-country evaluation and is retained as a design-boundary illustration, not as evidence: it is superseded by the main results (Section 7). The existing two-country pilot is retained because it clarifies a design boundary. Under the primary unconditioned prompt, the same model emits identical option probabilities when scored against USA and Mexico; only the external target distribution differs. Thus a matched-versus-cross comparison measures which adapter's *fixed* output distribution lies closer to each population's observed responses. It does not show that the adapter knows or conditions on its target country at inference.

In the committed pilot, the Mexico adapter has mean TVD 0.4882 on Mexico versus 0.5211 for the base model, while the USA adapter has 0.6377 on Mexico. The corresponding mean matched-minus-cross TVD difference is 0.1495 with a question-resampling 95% interval $[0.0964, 0.2152]$ across 35 items. On USA outcomes, however, the Mexico adapter is approximately neutral relative to base (0.3888 versus 0.3958), while the USA adapter degrades (0.5258). These values are verified transcriptions of the committed evaluation summaries.

The pilot therefore does not establish country specificity. A single fixed adapter can dominate another fixed adapter against either country, and $C = 2$ cannot reveal a stable country-by-policy cross-over pattern. The item bootstrap quantifies sensitivity to this 35-item battery, not uncertainty from training runs, model generation, respondents, or country sampling. The pilot motivates—rather than substitutes for—the multi-country placebo and operator-sensitivity design in the main paper.

B. Training configuration

Table 9 is the configuration used for every adapter in this paper. Any deviation is a declared sensitivity arm, not an ad hoc adjustment. Entries are transcribed from and verified against the canonical training artifacts under `DPO_train_test/`. The production data row refers to the sign(z) production bank (658 pairs, 526/132 split), which supersedes the legacy teacher-generator bank (≈3,594 triplets, 2,875 train) recorded in `DPO_train_test/`.

Table 9: Reference training configuration for the country-specific DPO adapters. All variants in the operator-sensitivity panel are declared relative to this configuration.

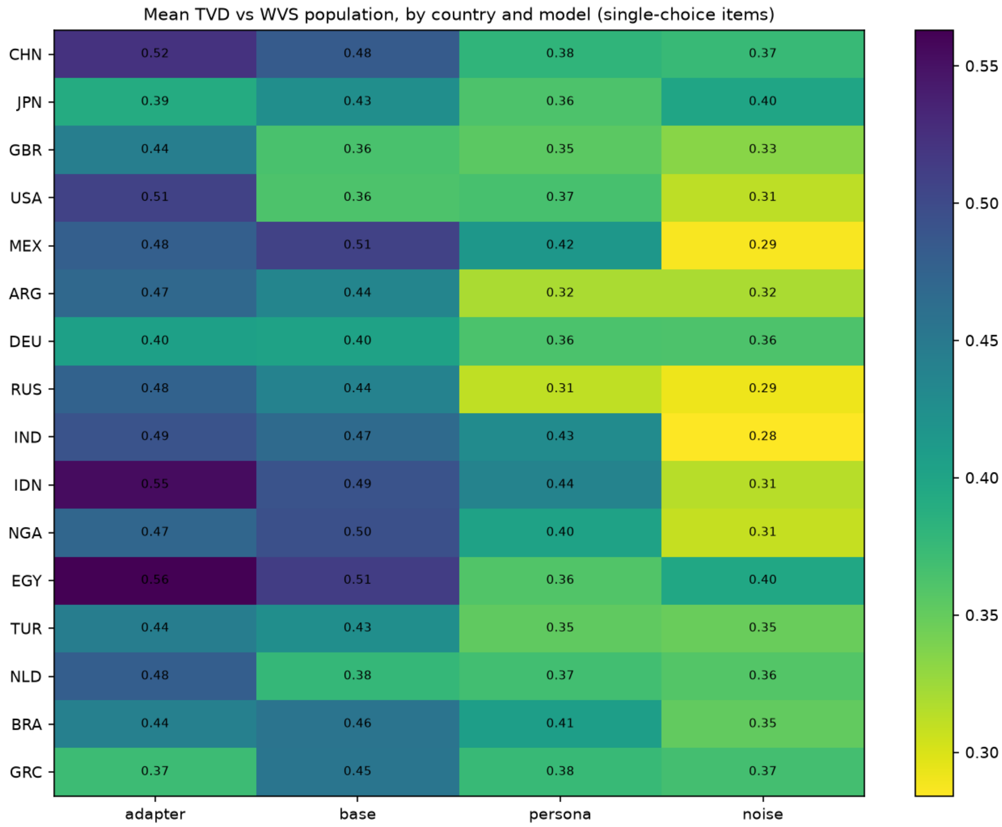
Component	Setting	Source/status
Base model	Llama-3.1-8B-Instruct (`meta-llama/Llama-3.1-8B-Instruct`)	`DPO_train.ipynb`
QLoRA quantization	4-bit NF4, double quant, fp16 compute	`DPO_train.ipynb`
LoRA rank/alpha	$r = 16$, $\alpha = 32$, dropout 0.05	`DPO_train.ipynb`
LoRA target modules	`q_proj`, `k_proj`, `v_proj`, `o_proj` (attention projections)	`DPO_train.ipynb`
DPO beta	$\beta = 0.1$	`DPO_train.ipynb`
Learning rate/schedule	1×10^{-4}, cosine, warmup ratio 0.03	`DPO_train.ipynb`
Epochs	1	`DPO_train.ipynb`
Batch size/gradient accumulation	per-device batch 1; 16 accumulation steps (effective 16)	`DPO_train.ipynb`
Synthetic triplets per country	526 of the shared 658-pair sign(z) bank; 132 held-out pairs per country (80/20, seed 42)	`CO2_RUN.md`, `manifest.json`
Hardware	Single Colab-class GPU (Tesla T4 in the committed run)	`DPO_train.ipynb`, `README.md`

C. Unrun operator-sensitivity panel

The following arms were declared before the sixteen-country evaluation and have not been executed. They are listed so a later release cannot quietly drop them. Compare the reference protocol with alternative scenario banks, alternate pair-generation materializations, deterministic-sign versus legacy-selector labels, quality-control threshold variations, a scorer-diagnostic-only arm, restricted anchor permutations that *retrain* adapters, and repeated DPO seeds. Each arm should report effects on synthetic-data diagnostics, learned-policy stability, and external transport. At least three independently materialized scenario/pair banks should be retained per condition. A remote API seed is not a sufficient reproducibility guarantee; materialized datasets are the primary artifacts.

D. Supplementary figures

The figures below are secondary views of the surfaces reported in the main text; they are generated by the same committed pipeline (`analysis/phase2/`) from the same artifacts.



	adapter	base	persona	noise
CHN	0.52	0.48	0.38	0.37
JPN	0.39	0.43	0.36	0.40
GBR	0.44	0.36	0.35	0.33
USA	0.51	0.36	0.37	0.31
MEX	0.48	0.51	0.42	0.29
ARG	0.47	0.44	0.32	0.32
DEU	0.40	0.40	0.36	0.36
RUS	0.48	0.44	0.31	0.29
IND	0.49	0.47	0.43	0.28
IDN	0.55	0.49	0.44	0.31
NGA	0.47	0.50	0.40	0.31
EGY	0.56	0.51	0.36	0.40
TUR	0.44	0.43	0.35	0.35
NLD	0.48	0.38	0.37	0.36
BRA	0.44	0.46	0.41	0.35
GRC	0.37	0.45	0.38	0.37

Figure S1: Total variation distance by country and model family, mean over the 30 single-choice items. Darker cells are further from the wvs population; noise is closest on average because the populations are diffuse.

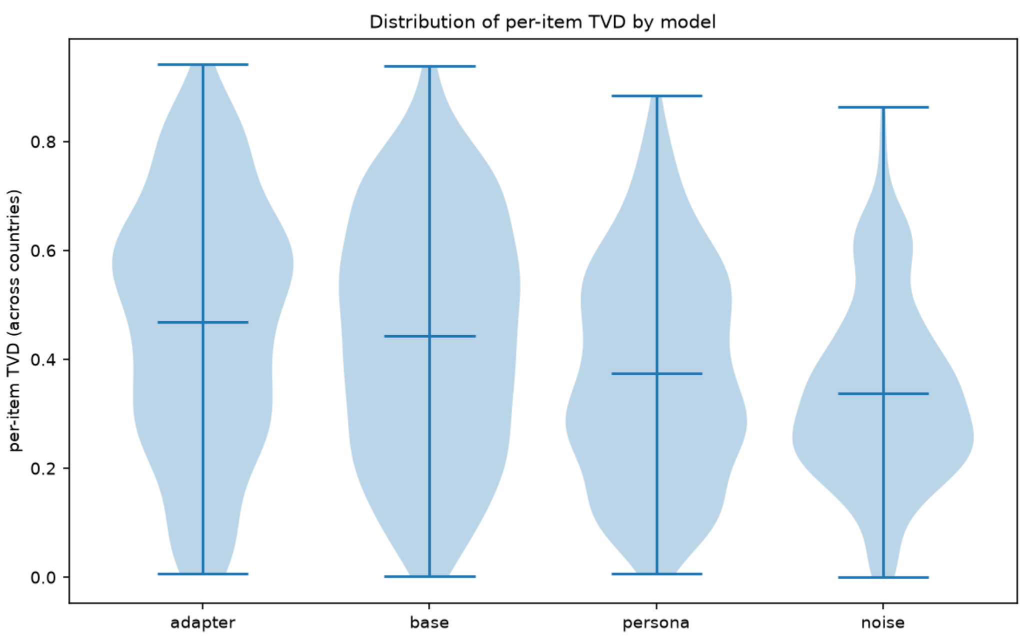


Figure S2: Per-item TVD distributions by model family. The adapters show the widest spread with a high median (over-concentration across items), persona is narrower, and noise has the lowest median and tightest spread.

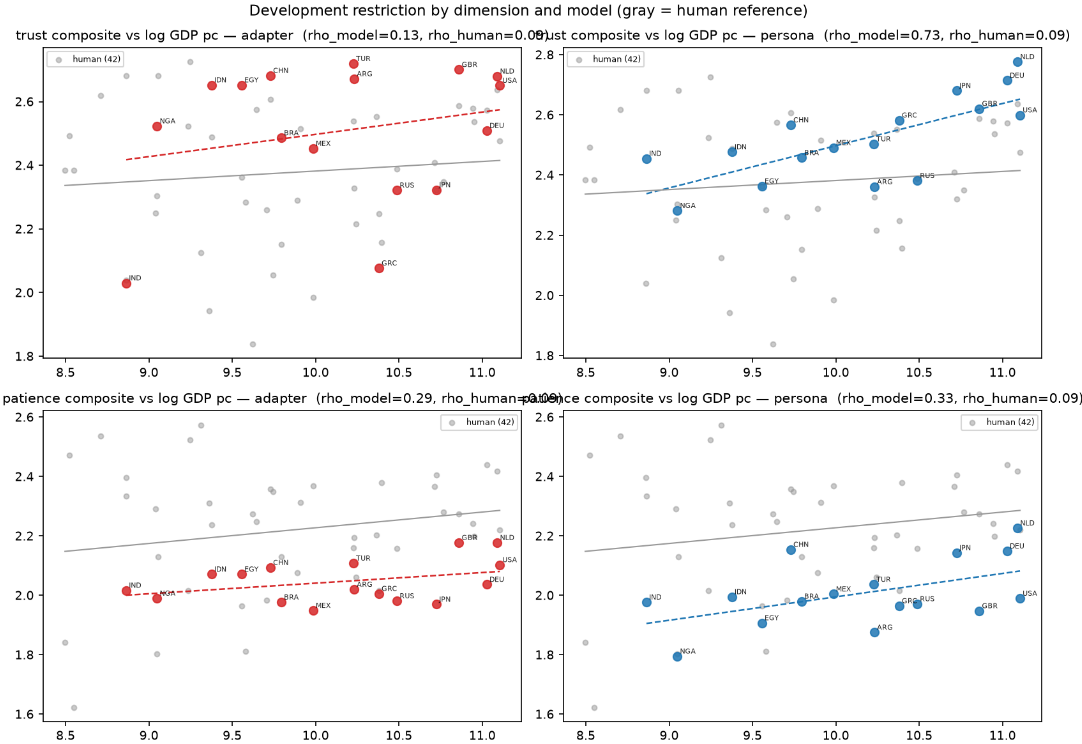


Figure S3: Development restriction: country composite versus log GDP per capita by dimension. Red points are adapters, blue points persona prompts, gray points the 42-country human overlay; Spearman ρ is reported in each panel title.

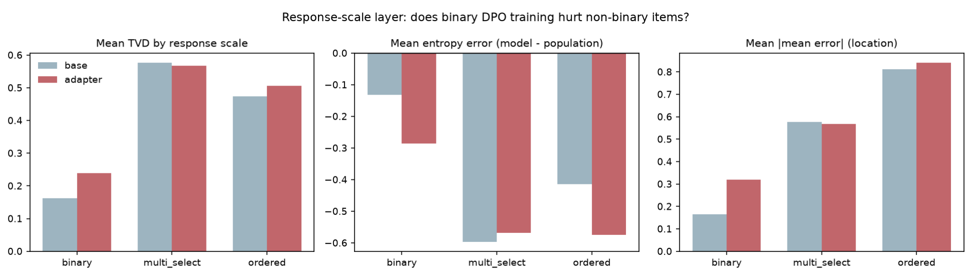


Figure S4: Inference-time temperature scaling: pooled TVD by temperature. Shape (dispersion) is recovered at $T = 3$ while top-option match is unchanged; location is unaffected. Reported as a diagnostic, not a training change.

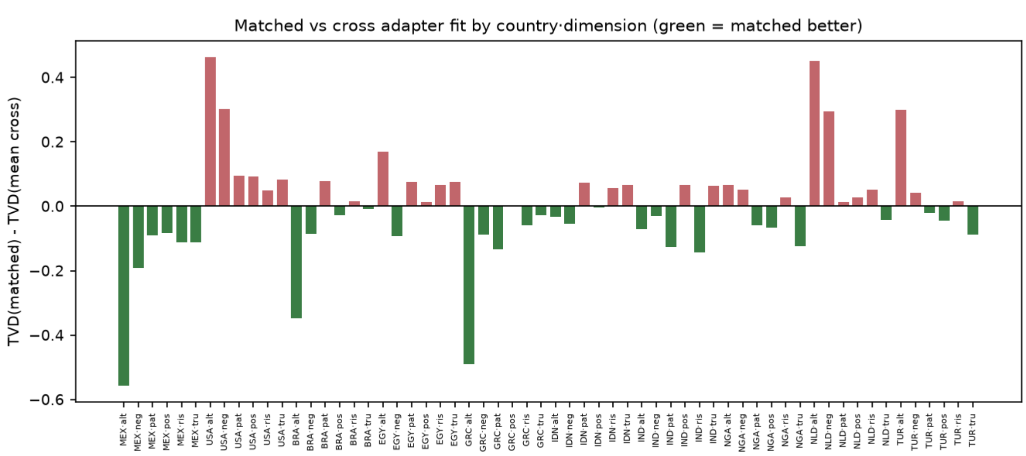


Figure S5: Matched-versus-cross TVD comparisons underlying the two-country pilot (Appendix A).

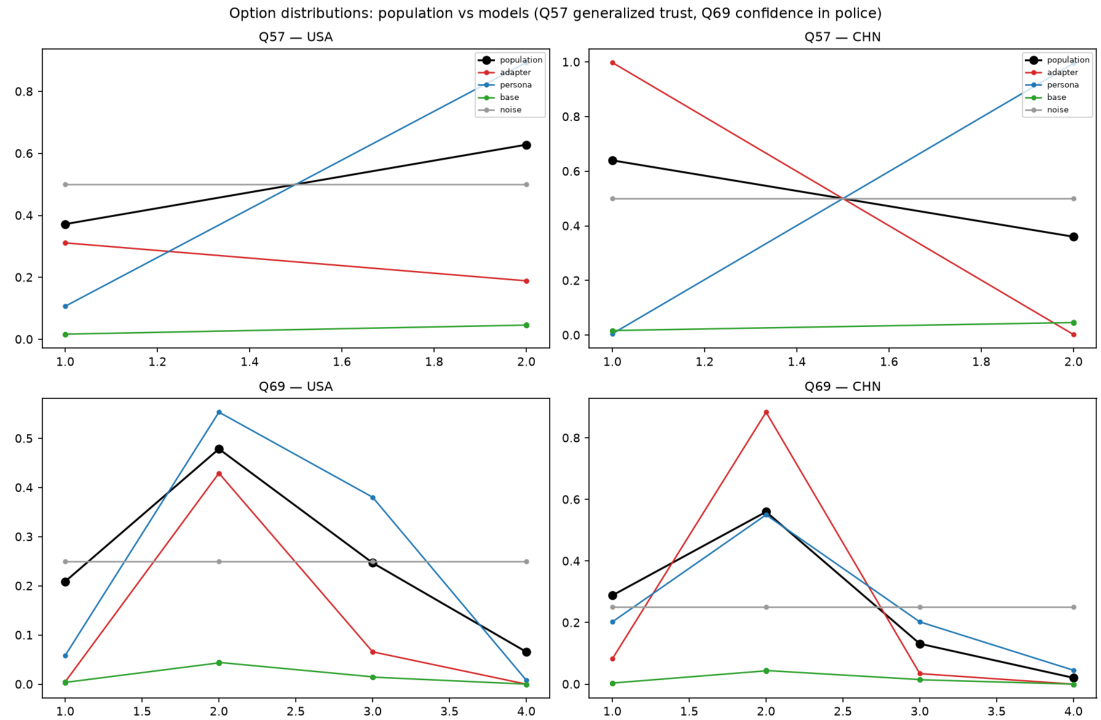


Figure S6: Item option distributions for two illustrative items (Q57 generalized trust; Q69 confidence in police). Black is the wvs population, red the adapter, blue the persona prompt, green the base, gray uniform noise.

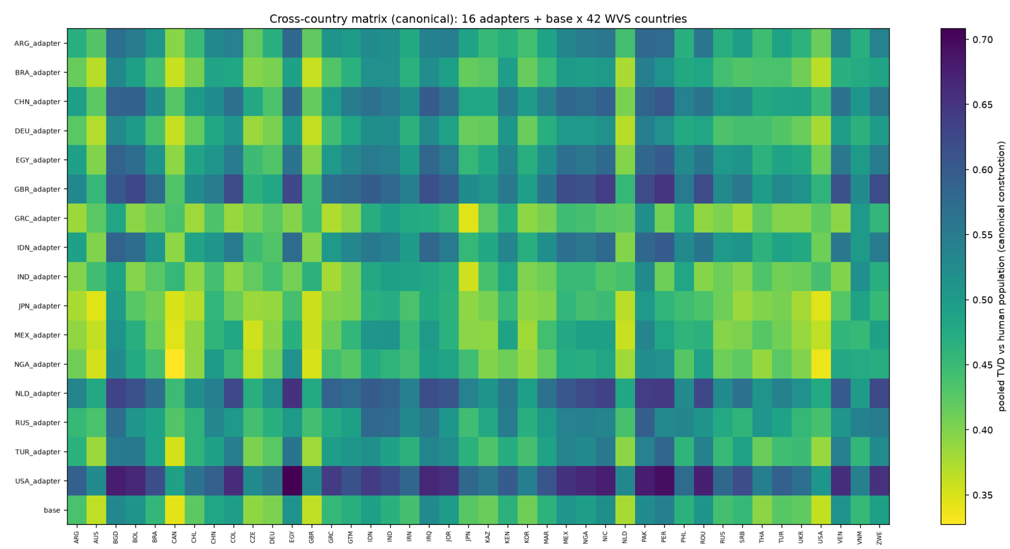


Figure S7: Full cross-country matrix: pooled TVD of each adapter and the base against all 42 WVS populations (canonical construction; own-country cells equal Table 1). Darker cells are further from the population. The row-wise minima are dominated by mid-scale populations, and no adapter is closest to its own country.

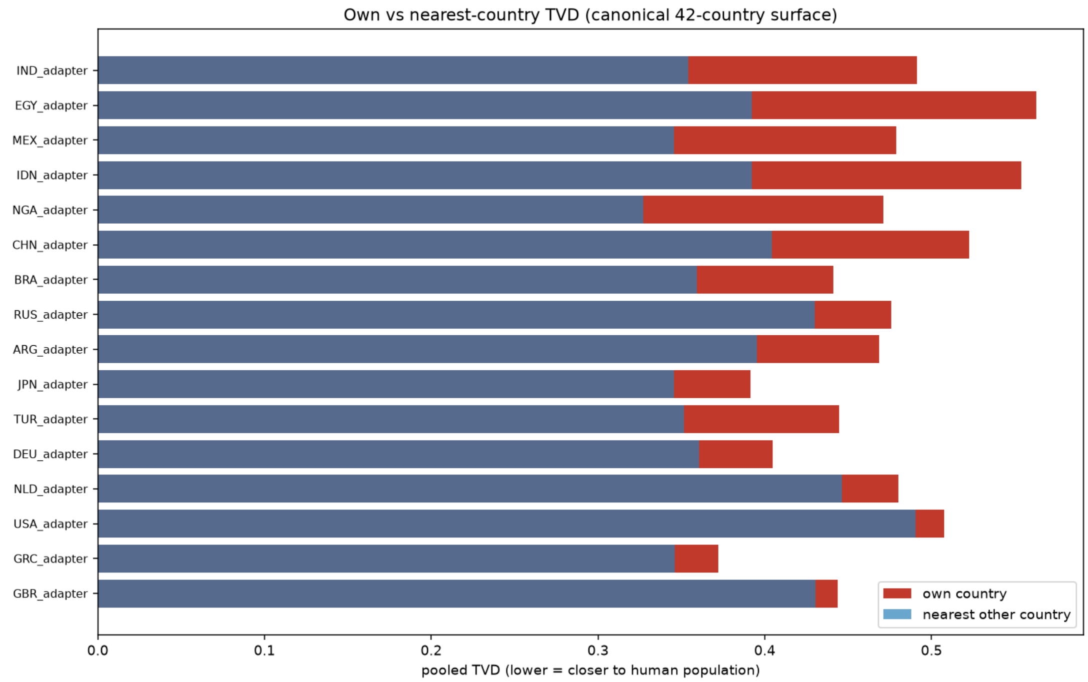


Figure S8: Own-country TVD versus the nearest other country for each adapter. Own-country bars (red) exceed the nearest-other bars (blue) for every adapter; the anti-specific set (NGA, IDN, MEX, EGY, IND) shows the largest gaps.

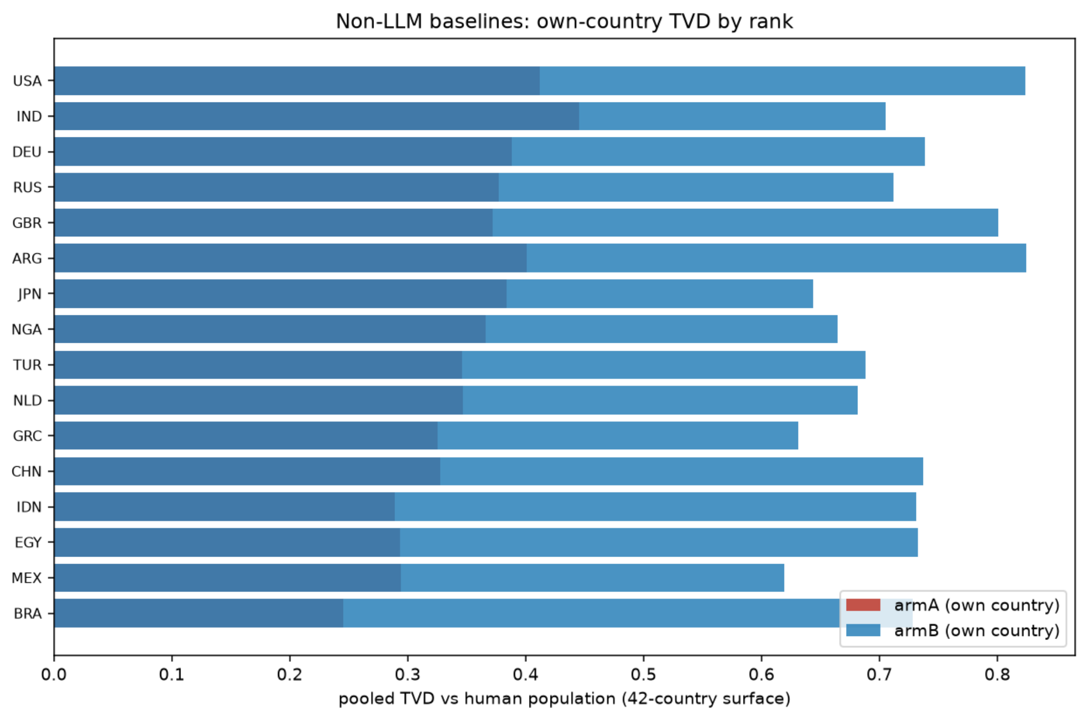


Figure S9: Non-LLM baseline arms on the 42-country location geometry: own-country TVD by own-rank. armA (maximum-entropy anchor transform) clusters near the flatness attractor (median rank 27.5 of 42); armB (sign-follower) ranks just below chance (14.0). Neither arm recovers its own country.

References

[1] L. P. Argyle, E. C. Busby, N. Fulda, J. R. Gubler, C. Rytting, and D. Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.

[2] A. Abels, E. Fernández Domingos, A. Shah, and T. Lenaerts. Improving cross-cultural survey simulation with calibrated value personas. arXiv:2605.16193, 2026. Under review at VALE 2026.

[3] J. Bisbee, J. D. Clinton, C. Dorff, B. Kenkel, and J. M. Larson. Synthetic replacements for human survey data? The perils of large language models. *Political Analysis*, 32(4):401–416, 2024.

[4] J. Boelaert, S. Coavoux, E. Ollion, I. Petev, and P. Präg. Machine bias. How do generative language models answer opinion polls? *Sociological Methods & Research*, 2025. DOI: `https://doi.org/10.1177/00491241251330582`.

[5] Y. Cao, H. Liu, A. Arora, I. Augenstein, P. Röttger, and D. Hershcovich. Specializing large language models to simulate survey response distributions for global populations. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics*, pages 3141–3154, 2025.

[6] C. M. Capra, A. Gonzalez-Bonorino, and F. Pantoja. LLMs can model non-WEIRD populations: Experiments with synthetic cultural agents. *Review of Experimental Economics*, forthcoming, 2025.

[7] D. T. Campbell and D. W. Fiske. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, 56(2):81–105, 1959.

[8] A. Gonzalez-Bonorino, K. Biriukova, and C. M. Capra. Measuring what we mean: Construct validity when measures are cheap. Working paper, EconLLM Lab, 2026.

[9] A. Dalloul and M. Pfeifer. Can LLMs mimic household surveys? From representative agents to population distributions. SSRN 6703798, 2026.

[10] E. Davidov, B. Meuleman, J. Cieciuch, P. Schmidt, and J. Billiet. Measurement equivalence in cross-national research. *Annual Review of Sociology*, 40:55–75, 2014.

[11] M. Dell and A. Rambachan. Estimation and inference with AI-generated data. NBER Summer Institute Methods Lecture, July 2026. Slides: `https://conference.nber.org/confer/2026/SI2026/ML/dell3.pdf`.

[12] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, 36, 2023.

[13] R. J. House, P. J. Hanges, M. Javidan, P. W. Dorfman, and V. Gupta, editors. *Culture, Leadership, and Organizations: The GLOBE Study of 62 Societies*. Sage Publications, 2004.

[14] A. Falk, A. Becker, T. Dohmen, B. Enke, D. Huffman, and U. Sunde. Global evidence on economic preferences. *Quarterly Journal of Economics*, 133(4):1645–1692, 2018.

[15] G. Hofstede, G. J. Hofstede, and M. Minkov. *Cultures and Organizations: Software of the Mind*, 3rd edition. McGraw-Hill, 2010.

[16] E. T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957.

[17] W. E. Deming and F. F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444, 1940.

[18] A. Golan, G. Judge, and D. Miller. *Maximum Entropy Econometrics: Robust Estimation with Limited Data*. Wiley, 1996.

[19] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

[20] R. Inglehart and C. Welzel. *Modernization, Cultural Change, and Democracy: The Human Development Sequence*. Cambridge University Press, 2005.

[21] S. Krsteski, G. Russo, S. Chang, R. West, and K. Gligorić. Valid survey simulations with limited human data: The roles of prompting, fine-tuning, and rectification. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 10887–10906, 2026.

[22] J. Ludwig, S. Mullainathan, and A. Rambachan. Large language models: An applied econometric framework. *Annual Review of Economics*, 18:283–316, 2026.

[23] M. Muthukrishna, A. V. Bell, J. Henrich, C. M. Curtin, A. Gedranovich, J. McInerney, and B. Thue. Beyond Western, educated, industrial, rich, and democratic (WEIRD) psychology: Measuring and mapping scales of cultural and psychological distance. *Psychological Science*, 31(6):678–701, 2020.

[24] R. Rafailov, A. Sharma, E. Mitchell, S. Ermon, C. D. Manning, and C. Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, 36, 2023.

[25] S. H. Schwartz. Universals in the content and structure of values: Theoretical advances and empirical tests in 20 countries. In M. P. Zanna, editor, *Advances in Experimental Social Psychology*, volume 25, pages 1–65. Academic Press, 1992.

[26] A. Sharifnassab, S. Salehkaleybar, S. Ghiassian, S. Kanoria, and D. Schuurmans. Soft preference optimization: Aligning language models to expert distributions. arXiv:2405.00747, 2024.

[27] J.-B. E. M. Steenkamp and H. Baumgartner. Assessing measurement invariance in cross-national consumer research. *Journal of Consumer Research*, 25(1):78–90, 1998.

[28] J. Suh, E. Jahanparast, S. Moon, M. Kang, and S. Chang. Language model fine-tuning on scaled survey data for predicting distributions of public opinions. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. ACL anthology 2025.acl-long.1028.

[29] Y. Tao, O. Viberg, R. S. Baker, and R. F. Kizilcec. Cultural bias and cultural alignment of large language models. *PNAS Nexus*, 3(9):pgae346, 2024.

[30] A. Wang, J. Morgenstern, and J. P. Dickerson. Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence*, 7:400–411, 2025.

[31] World Values Survey Association. World Values Survey Wave 7 (2017–2022). DOI: `https://doi.org/10.14281/18241.18`, 2022.